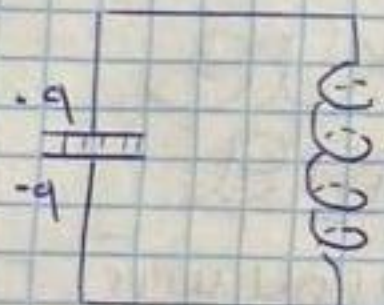
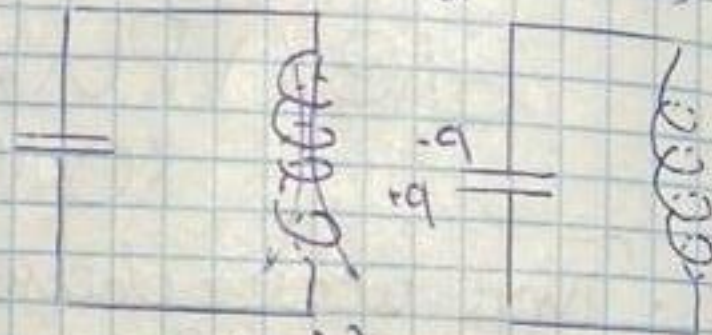


Свободные колеб. заряда
в идеальной контуре ($R=0$)



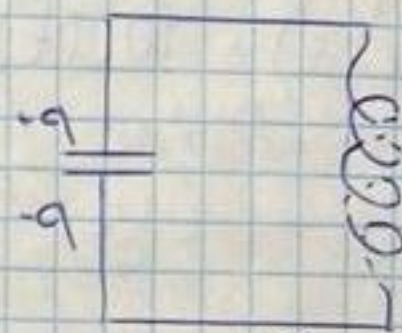
$$W = \frac{q^2}{2C}$$



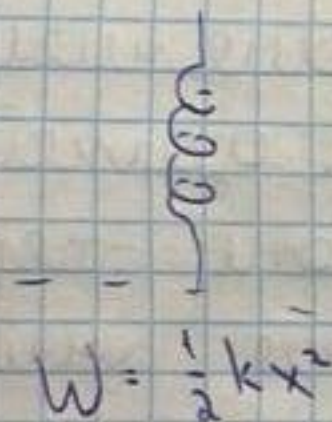
$$W = \frac{L i^2}{2}$$



$$W = \frac{q^2}{2C}$$



$$W = \frac{L i^2}{2}$$



$$W = \frac{1}{2} k x^2$$



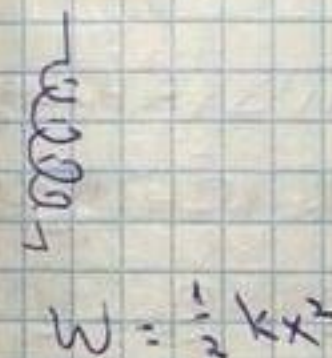
$$W = \frac{1}{2} m v^2$$



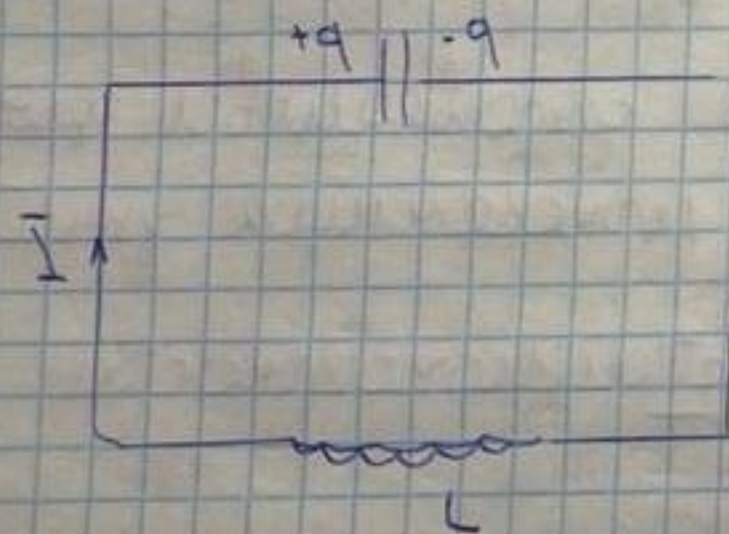
$$W = \frac{1}{2} k x^2$$



$$W = \frac{1}{2} m v^2$$



$$W = \frac{1}{2} k x^2$$



$$\begin{aligned} IR &= \varphi_1 - \varphi_2 + \mathcal{E}_{12} \\ R=0, \varphi_1 - \varphi_2 &= -q/C \\ \mathcal{E}_{12} &= -L \frac{dI}{dt} \\ L \frac{dI}{dt} + q/C &= 0 \end{aligned}$$

$$\ddot{q} + \frac{q}{LC} = 0, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\ddot{q} + \omega_0^2 q = 0$$

$$q = q_m \cos(\omega_0 t + \alpha)$$

$$U = \frac{q_m^2}{2C} \cos^2(\omega_0 t + \alpha) = U_m \cos^2(\omega_0 t + \alpha)$$

$$I = -\omega_0 q_m \sin(\omega t + \alpha) = I_m \cos(\omega t + \alpha + \frac{\pi}{2})$$

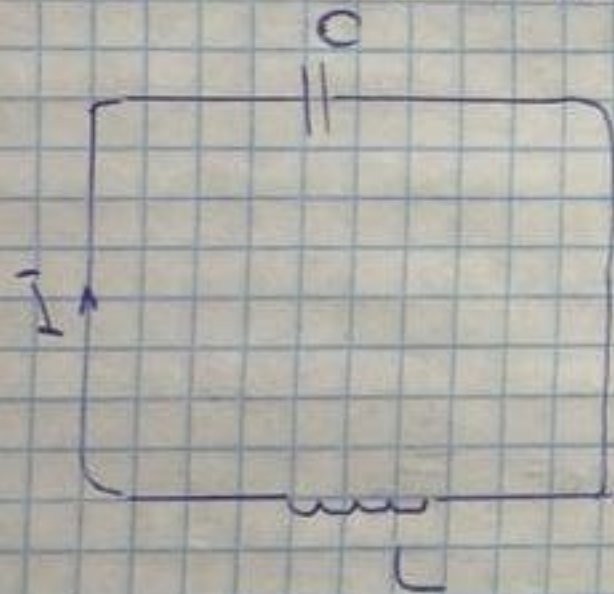
$$W_{max} = W_C(t) + W_L(t)$$

$$W_{Cmax}$$

$$W_{Lmax}$$

$$U_m = \sqrt{\frac{L}{C}} I_m$$

Свободные затухающие колебания в контуре



$$IR = -\frac{q}{C} - L \frac{dI}{dt}$$

$$q'' + \frac{R}{L} q' + \frac{1}{LC} q = 0$$

$$\beta = \frac{R}{2L} \text{ — коэффициент затухания}$$

$$q'' + 2\beta q' + \omega_0^2 q = 0$$

$$q = q_m e^{-\beta t} \cos(\omega t + \alpha) \text{ — амплитуда}$$

заряд зависит от времени

$$\omega = \sqrt{\omega_0^2 - \beta^2} = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$U = U_m e^{-\beta t} \cos(\omega t + \alpha)$$

$$I = q_m e^{-\beta t} (-\beta \cos(\omega t + \alpha) - \omega \sin(\omega t + \alpha))$$

$$I = \omega_0 q_m e^{-\beta t} \cos(\omega t + \alpha + \varphi)$$

$$\cos \varphi = -\frac{\beta}{\sqrt{\omega^2 + \beta^2}}, \sin \varphi = \frac{\omega}{\sqrt{\omega^2 + \beta^2}} = \frac{\omega}{\omega_0} = \frac{\omega}{\beta/\omega_0}$$

Логарифмический декремент затухания обратен числу колебаний, соверш. за время, в теч. кот. амплитуда уменьш. в е раз.

$$\lambda = \ln \frac{A(t)}{A(t+\tau)} = \beta \tau$$

